

Wk10-Q09 Timer/Counter: PC ____ รหัส ____ ชื่อ-สกุล ____

เขียนโปรแกรมด้วย Computer Lab(แบบไม่ต่อเน็ต) เพื่อทำงานตามภารกิจที่กำหนด

Mission Y: <https://shorturl.at/dZxag>

- ต่อ MAX7219 Display กับ MCS51 ทดสอบการทำงาน
- แล้วปรับให้นับ Pulse ด้วย Counter-0 ที่ส่งมาจาก ESP32

Y. MAX7219 Display DPTR with STC89C52 Board

1. Display on MAX7219 by STC89C52

```
=====
;
; Test MAX7219 7 Segment Scan Display Control
; Compiler: ASEM51.EXE
; Hardware: STC89C52 Board
; Created: 12 Feb 2026
; By: Wichai Srisuruk @SUT-Microprocessor Lab
;
=====

Pin_LED30 EQU P3.0
Pin_LED31 EQU P3.1

pin_DI EQU P1.7
pin_CS EQU P1.6
pin_CK EQU P1.5

ORG 0000H
JMP 0100H

ORG 0100H
CLR EA
MOV SP,#2FH
SETB pin_DI
SETB pin_CS
SETB pin_CK
CALL myDelay
CALL Initial_MAX7219_Seg
CALL MAX7219_Display_Test

MOV DPTR,#4567
LOOP: CALL Display_BCD2MAX7219
INC DPTR
CALL myDelay
JMP LOOP

;
=====
; Input: DPTR = 16-bit Hex value
; Output: R6, R5, R4 = BCD result (Packed)
DPTR_HEX2BCD:
MOV R4, #0
MOV R5, #0
MOV R6, #0
MOV R7, #16 ; Loop counter (16 bits to shift)
_CONV_LOOP:
XCH A, DPL
ADD A, ACC
XCH A, DPL
XCH A, DPH
ADDC A, ACC
XCH A, DPH
; Process R4 (Tens and Ones)
MOV A, R4
ADDC A, ACC
DA A
MOV R4, A
; Process R5 (Thousands and Hundreds)
MOV A, R5
ADDC A, ACC
DA A
MOV R5, A
; Process R6 (Ten-thousands)
MOV A, R6
ADDC A, ACC
DA A
MOV R6, A
; Repeat 16 times
DJNZ R7, _CONV_LOOP
RET

;
=====
Display_BCD2MAX7219:
PUSH DPH
PUSH DPL
CALL DPTR_HEX2BCD
MOV R7, #0
CALL Display_R7654
POP DPL
POP DPH
RET
```

```

;=====
Display_R7654:
    MOV     B,#8
    MOV     A,R7
    CALL    _Crack_Dis01
    MOV     A,R6
    CALL    _Crack_Dis01
    MOV     A,R5
    CALL    _Crack_Dis01
    MOV     A,R4
_Crack_Dis01:
    PUSH    Acc
    SWAP    A
    CALL    _Crack_Dis00
    POP     Acc
_Crack_Dis00:
    ANL     A,#0FH
    MOV     DPTR,#Segment_Table
    MOVC    A,@A+DPTR
    MOV     DPL,A
    MOV     DPH,B
    CALL    Send_DPTR2MAX7219
    DEC     B
    RET

;=====
Initial_MAX7219_Seg:
    MOV     DPTR,#0B07H; // B=Scan {0-7 Position}
    CALL    Send_DPTR2MAX7219
    MOV     DPTR,#0900H; // 9=Mode {00=NoCode,ff=code}
    CALL    Send_DPTR2MAX7219
    MOV     DPTR,#0F00H; // F=Test {0,1}
    CALL    Send_DPTR2MAX7219
    MOV     DPTR,#0A08H; // A=Intensity = 8/15
    CALL    Send_DPTR2MAX7219
    MOV     DPTR,#0C01H; // c=shutdown {0,1}
    CALL    Send_DPTR2MAX7219
    RET

MAX7219_Display_Test:
    MOV     DPTR,#0F01H ;0F01 : Display Test
    CALL    Send_DPTR2MAX7219
    CALL    myDelay
    MOV     DPTR,#0F00H ;0F00 : Display Normal
    CALL    Send_DPTR2MAX7219
    RET

Send_DPTR2MAX7219:
    SETB    pin_DI
    SETB    pin_CK
    CLR     pin_CS
    DB      0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
    MOV     A,DPH
    CALL    _Send_DPTR2MAX7219_8Bit
    MOV     A,DPL
    CALL    _Send_DPTR2MAX7219_8Bit
    DB      0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
    SETB    pin_DI
    SETB    pin_CK
    SETB    pin_CS
    RET

_Send_DPTR2MAX7219_8Bit:
    MOV     R7,#8
_ShiftBit_DPTR2MAX7219:
    RLC     A
    MOV     pin_DI,C
    DB      0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
    CLR     pin_CK
    DB      0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
    SETB    pin_CK
    DJNZ    R7,_ShiftBit_DPTR2MAX7219
    RET

Segment_Table:
; TABCDEFG
    DB 7EH, 30H, 6DH, 79H ; 0, 1, 2, 3
    DB 33H, 5BH, 5FH, 70H ; 4, 5, 6, 7
    DB 7FH, 7BH, 77H, 1FH ; 8, 9, A, B
    DB 4EH, 3DH, 4FH, 47H ; C, D, E, F
    DB 00H, 01H ; 10: Blank, 11: Minus

;=====
myDelay:CLR A
    MOV     B,A
    MOV     R7,#5
_DLY00:
    DJNZ    Acc,_DLY00
    DJNZ    B,_DLY00
    DJNZ    R7,_DLY00
    RET

END

```

2. ESP32 Send Pulse

```
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(19, INPUT_PULLUP);
  pinMode(21, INPUT_PULLUP);
  pinMode(22, INPUT_PULLUP);
  pinMode(23, INPUT_PULLUP);
}

void SendPulse(int nPulse, int xDelay) {
  digitalWrite(LED_BUILTIN, LOW);
  for (int i = 0; i < nPulse; i++) {
    digitalWrite(5, LOW);
    delayMicroseconds(xDelay);
    digitalWrite(5, HIGH);
    delayMicroseconds(xDelay);
  }
  delay(3000);
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH);
  if (digitalRead(19) == LOW) SendPulse(10, 400);
  if (digitalRead(21) == LOW) SendPulse(100, 300);
  if (digitalRead(22) == LOW) SendPulse(1000, 200);
  if (digitalRead(23) == LOW) SendPulse(1111, 100);
}
```

ESP32_Send.ino

```
1  void setup() {
2    pinMode(LED_BUILTIN, OUTPUT);
3    pinMode(5, OUTPUT);
4    pinMode(19, INPUT_PULLUP);
5    pinMode(21, INPUT_PULLUP);
6    pinMode(22, INPUT_PULLUP);
7    pinMode(23, INPUT_PULLUP);
8  }
9
10 void SendPulse(int nPulse, int xDelay) {
11   digitalWrite(LED_BUILTIN, LOW);
12   for (int i = 0; i < nPulse; i++) {
13     digitalWrite(5, LOW);
14     delayMicroseconds(xDelay);
15     digitalWrite(5, HIGH);
16     delayMicroseconds(xDelay);
17   }
18   delay(3000);
19 }
20
21 void loop() {
22   digitalWrite(LED_BUILTIN, HIGH);
23   if (digitalRead(19) == LOW) SendPulse(10, 400);
24   if (digitalRead(21) == LOW) SendPulse(100, 300);
25   if (digitalRead(22) == LOW) SendPulse(1000, 200);
26   if (digitalRead(23) == LOW) SendPulse(1111, 100);
27 }
28
```